

Gudipatoju Nitesh

AI / ML Engineer | Final-Year B.Tech (AIML)

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PROFESSIONAL SUMMARY

AI/ML Engineer with proven results: **~94% accuracy** on CNN-based image classification, **~90% accuracy** on agro-meteorological prediction, and **~40% improvement** in chatbot response relevance via ChromaDB RAG pipeline. Skilled in end-to-end delivery — model training, feature engineering, FastAPI backend design, and Docker deployment. Seeking an AI/ML internship to drive real-world impact at scale.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript, SQL
AI / ML:	TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, NLP, Hugging Face, LangChain, OpenCV
LLM & Vector DB:	RAG, ChromaDB, Prompt Engineering
Backend & DevOps:	FastAPI, Docker, REST APIs, JWT, Git, Streamlit, Jupyter

PROJECTS

AI Chatbot with Personality | LLM · Whisper · ChromaDB · FastAPI · Docker ↗ *Nov 2025 – Feb 2026*

- Improved chatbot response relevance by ~40% using ChromaDB vector similarity search for persistent contextual memory.
- Built Whisper speech-to-text pipeline; designed 5 JWT-secured FastAPI endpoints; containerised with Docker.

AI-Based Smart Irrigation System | Random Forest · XGBoost · Gradient Boosting ↗ *Jan 2026 – Present*

- Achieved ~90% classification accuracy predicting irrigation actions from temperature, humidity, and rainfall data.
- Compared 3 ensemble models via cross-validated F1; XGBoost outperformed baseline by ~15% on unseen test data.
- Added AI crop advisory module delivering plain-language guidance; estimated to reduce water usage by ~20%.

Hair Disease Detection | CNN · OpenCV · Deep Learning *2025*

- Attained ~94% prediction accuracy on scalp-image datasets (alopecia, dandruff, psoriasis) using a custom CNN.
- Applied OpenCV augmentation to handle class imbalance; reduced misclassification rate by ~18% vs baseline.
- Pipeline designed for web/mobile health deployment; inference runs under 200ms per image.

Placement Intelligence Platform | Python · Random Forest · Streamlit ↗ *Aug 2025 – Jan 2026*

- Predicted student placement probability at ~82% accuracy using 5-fold cross-validated Random Forest.
- Engineered academic & experience features to reduce overfitting; built live Streamlit scoring dashboard.

INDUSTRY EXPERIENCE

Data Analytics Job Simulation – Deloitte *Sep – Oct 2025*

Forensic data analysis and insight reporting on real-world client-style datasets; sharpened analytical thinking and communication.

GenAI-Powered Data Analytics Simulation – Tata Group *Sep 2025*

Applied EDA, AI-based prediction, and business data storytelling; developed risk profiling and insight-driven decision-making.

CERTIFICATIONS

Introduction to Generative AI Studio – Google Cloud

LLM workflows, prompt design, and cloud-based AI deployment.

Career Essentials in Data Analysis – Microsoft & LinkedIn

Data analysis, visualisation, and analytics best practices.

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning *2022 – 2026*

Nalla Malla Reddy Engineering College, Hyderabad

Intermediate (MPC) *2020 – 2022*

Ekalavya Junior College, Kothapet

Secondary School Certificate (SSC) *2019 – 2020*

Little Flower High School, L.B. Nagar